In [1]:

```python
import numpy as np
import matplotlib.pyplot as plt

import torch
import torch.nn as nn
from torch.utils.data import DataLoader
from torchvision import datasets, transforms

transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Lambda(lambda x: x.view(-1))
])

train_dataset = datasets.FashionMNIST(
    root="./data",
    train=True,
    download=True,
    transform=transform
)

test_dataset = datasets.FashionMNIST(
    root="./data",
    train=False,
    download=True,
    transform=transform
)

train_loader = DataLoader(train_dataset, batch_size=100, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=100, shuffle=False)

class Model1(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc1 = nn.Linear(784, 128)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(128, 10)

    def forward(self, x):
        x = self.fc1(x)
        x = self.relu(x)
        x = self.fc2(x)
        return x

class Model2(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc1 = nn.Linear(784, 48)
        self.relu = nn.ReLU()
        self.dropout = nn.Dropout(p=0.2)
        self.fc2 = nn.Linear(48, 10)

    def forward(self, x):
        x = self.fc1(x)
        x = self.relu(x)
        x = self.dropout(x)
        x = self.fc2(x)
        return x
```

```python
def train_one_epoch(model, loader, criterion, optimizer):
    model.train()

    total_loss = 0.0
    correct = 0
    total = 0

    for x, y in loader:
        optimizer.zero_grad()
        outputs = model(x)
        loss = criterion(outputs, y)
        loss.backward()
        optimizer.step()

        total_loss += loss.item() * x.size(0)
        _, predicted = torch.max(outputs, 1)
        correct += (predicted == y).sum().item()
        total += y.size(0)

    avg_loss = total_loss / total
    accuracy = correct / total
    return avg_loss, accuracy

def evaluate(model, loader, criterion):
    model.eval()

    total_loss = 0.0
    correct = 0
    total = 0

    with torch.no_grad():
        for x, y in loader:
            outputs = model(x)
            loss = criterion(outputs, y)

            total_loss += loss.item() * x.size(0)
            _, predicted = torch.max(outputs, 1)
            correct += (predicted == y).sum().item()
            total += y.size(0)

    avg_loss = total_loss / total
    accuracy = correct / total
    return avg_loss, accuracy

num_epochs = 40
criterion = nn.CrossEntropyLoss()

model1 = Model1()
optimizer1 = torch.optim.SGD(model1.parameters(), lr=0.1, weight_decay=0.0)

model1_train_losses = []
model1_test_losses = []
model1_train_accs = []
model1_test_accs = []

for epoch in range(num_epochs):
    train_loss, train_acc = train_one_epoch(model1, train_loader, criterion, opt
    test_loss, test_acc = evaluate(model1, test_loader, criterion)

    model1_train_losses.append(train_loss)
```

```
    model1_test_losses.append(test_loss)
    model1_train_accs.append(train_acc)
    model1_test_accs.append(test_acc)

    print(f"Model 1 - Epoch {epoch+1}/{num_epochs}")
    print(f"Train Loss: {train_loss:.4f}, Train Acc: {train_acc:.4f}")
    print(f"Test  Loss: {test_loss:.4f}, Test  Acc: {test_acc:.4f}")
    print("-" * 40)

model2 = Model2()
optimizer2 = torch.optim.SGD(model2.parameters(), lr=0.1, weight_decay=0.0001)

model2_train_losses = []
model2_test_losses = []
model2_train_accs = []
model2_test_accs = []

for epoch in range(num_epochs):
    train_loss, train_acc = train_one_epoch(model2, train_loader, criterion, opt
    test_loss, test_acc = evaluate(model2, test_loader, criterion)

    model2_train_losses.append(train_loss)
    model2_test_losses.append(test_loss)
    model2_train_accs.append(train_acc)
    model2_test_accs.append(test_acc)

    print(f"Model 2 - Epoch {epoch+1}/{num_epochs}")
    print(f"Train Loss: {train_loss:.4f}, Train Acc: {train_acc:.4f}")
    print(f"Test  Loss: {test_loss:.4f}, Test  Acc: {test_acc:.4f}")
    print("-" * 40)

epochs = range(1, num_epochs + 1)


w1_fc1 = model1.fc1.weight.detach().cpu().numpy().flatten()
w1_fc2 = model1.fc2.weight.detach().cpu().numpy().flatten()

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(w1_fc1, bins=50)
ax.set_title("Model 1 - Input Layer Weights")
ax.set_xlabel("Weight Value")
ax.set_ylabel("Frequency")
plt.show()
plt.close(fig)

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(w1_fc2, bins=50)
ax.set_title("Model 1 - Hidden/Output Layer Weights")
ax.set_xlabel("Weight Value")
ax.set_ylabel("Frequency")
plt.show()
plt.close(fig)

w2_fc1 = model2.fc1.weight.detach().cpu().numpy().flatten()
w2_fc2 = model2.fc2.weight.detach().cpu().numpy().flatten()

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(w2_fc1, bins=50)
ax.set_title("Model 2 - Input Layer Weights")
ax.set_xlabel("Weight Value")
```

```
ax.set_ylabel("Frequency")
plt.show()
plt.close(fig)

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(w2_fc2, bins=50)
ax.set_title("Model 2 - Hidden/Output Layer Weights")
ax.set_xlabel("Weight Value")
ax.set_ylabel("Frequency")
plt.show()
plt.close(fig)
```

```
Model 1 - Epoch 1/40
Train Loss: 0.6901, Train Acc: 0.7649
Test  Loss: 0.5833, Test  Acc: 0.7962
----------------------------------------
Model 1 - Epoch 2/40
Train Loss: 0.4728, Train Acc: 0.8323
Test  Loss: 0.4766, Test  Acc: 0.8350
----------------------------------------
Model 1 - Epoch 3/40
Train Loss: 0.4241, Train Acc: 0.8489
Test  Loss: 0.4299, Test  Acc: 0.8478
----------------------------------------
Model 1 - Epoch 4/40
Train Loss: 0.3958, Train Acc: 0.8581
Test  Loss: 0.4204, Test  Acc: 0.8483
----------------------------------------
Model 1 - Epoch 5/40
Train Loss: 0.3764, Train Acc: 0.8659
Test  Loss: 0.4117, Test  Acc: 0.8528
----------------------------------------
Model 1 - Epoch 6/40
Train Loss: 0.3606, Train Acc: 0.8720
Test  Loss: 0.4123, Test  Acc: 0.8517
----------------------------------------
Model 1 - Epoch 7/40
Train Loss: 0.3467, Train Acc: 0.8750
Test  Loss: 0.3782, Test  Acc: 0.8650
----------------------------------------
Model 1 - Epoch 8/40
Train Loss: 0.3347, Train Acc: 0.8793
Test  Loss: 0.3725, Test  Acc: 0.8655
----------------------------------------
Model 1 - Epoch 9/40
Train Loss: 0.3269, Train Acc: 0.8814
Test  Loss: 0.3632, Test  Acc: 0.8693
----------------------------------------
Model 1 - Epoch 10/40
Train Loss: 0.3167, Train Acc: 0.8847
Test  Loss: 0.3650, Test  Acc: 0.8685
----------------------------------------
Model 1 - Epoch 11/40
Train Loss: 0.3097, Train Acc: 0.8861
Test  Loss: 0.3667, Test  Acc: 0.8683
----------------------------------------
Model 1 - Epoch 12/40
Train Loss: 0.3025, Train Acc: 0.8897
Test  Loss: 0.3568, Test  Acc: 0.8702
----------------------------------------
Model 1 - Epoch 13/40
Train Loss: 0.2934, Train Acc: 0.8935
Test  Loss: 0.3683, Test  Acc: 0.8642
----------------------------------------
Model 1 - Epoch 14/40
Train Loss: 0.2889, Train Acc: 0.8952
Test  Loss: 0.3444, Test  Acc: 0.8754
----------------------------------------
Model 1 - Epoch 15/40
Train Loss: 0.2819, Train Acc: 0.8973
Test  Loss: 0.3596, Test  Acc: 0.8710
----------------------------------------
```

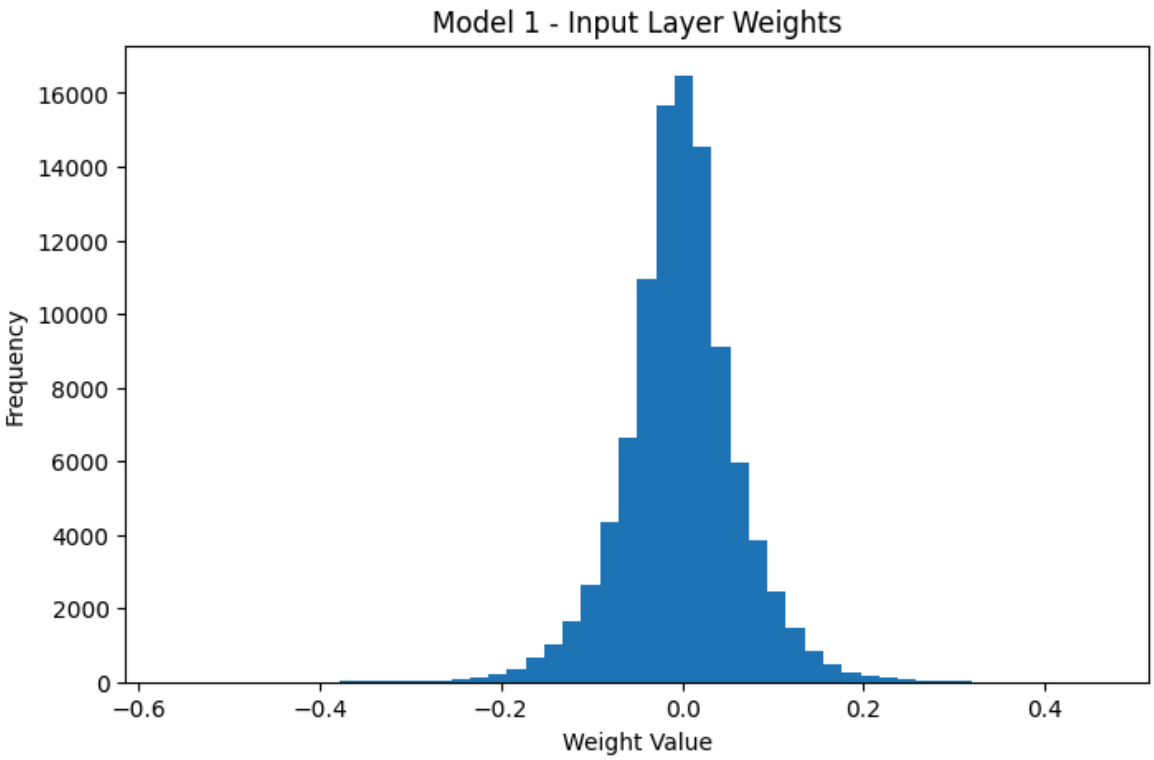
```
Model 1 - Epoch 16/40
Train Loss: 0.2773, Train Acc: 0.8988
Test  Loss: 0.3553, Test  Acc: 0.8683
----------------------------------------
Model 1 - Epoch 17/40
Train Loss: 0.2724, Train Acc: 0.9012
Test  Loss: 0.3358, Test  Acc: 0.8785
----------------------------------------
Model 1 - Epoch 18/40
Train Loss: 0.2672, Train Acc: 0.9018
Test  Loss: 0.3723, Test  Acc: 0.8621
----------------------------------------
Model 1 - Epoch 19/40
Train Loss: 0.2633, Train Acc: 0.9034
Test  Loss: 0.3598, Test  Acc: 0.8727
----------------------------------------
Model 1 - Epoch 20/40
Train Loss: 0.2588, Train Acc: 0.9065
Test  Loss: 0.3391, Test  Acc: 0.8800
----------------------------------------
Model 1 - Epoch 21/40
Train Loss: 0.2549, Train Acc: 0.9081
Test  Loss: 0.3359, Test  Acc: 0.8798
----------------------------------------
Model 1 - Epoch 22/40
Train Loss: 0.2498, Train Acc: 0.9089
Test  Loss: 0.3291, Test  Acc: 0.8821
----------------------------------------
Model 1 - Epoch 23/40
Train Loss: 0.2469, Train Acc: 0.9092
Test  Loss: 0.3311, Test  Acc: 0.8844
----------------------------------------
Model 1 - Epoch 24/40
Train Loss: 0.2435, Train Acc: 0.9110
Test  Loss: 0.3423, Test  Acc: 0.8757
----------------------------------------
Model 1 - Epoch 25/40
Train Loss: 0.2380, Train Acc: 0.9139
Test  Loss: 0.3283, Test  Acc: 0.8861
----------------------------------------
Model 1 - Epoch 26/40
Train Loss: 0.2345, Train Acc: 0.9141
Test  Loss: 0.3286, Test  Acc: 0.8830
----------------------------------------
Model 1 - Epoch 27/40
Train Loss: 0.2333, Train Acc: 0.9137
Test  Loss: 0.3352, Test  Acc: 0.8832
----------------------------------------
Model 1 - Epoch 28/40
Train Loss: 0.2292, Train Acc: 0.9154
Test  Loss: 0.3338, Test  Acc: 0.8820
----------------------------------------
Model 1 - Epoch 29/40
Train Loss: 0.2254, Train Acc: 0.9176
Test  Loss: 0.3207, Test  Acc: 0.8872
----------------------------------------
Model 1 - Epoch 30/40
Train Loss: 0.2215, Train Acc: 0.9192
Test  Loss: 0.3330, Test  Acc: 0.8819
----------------------------------------
```

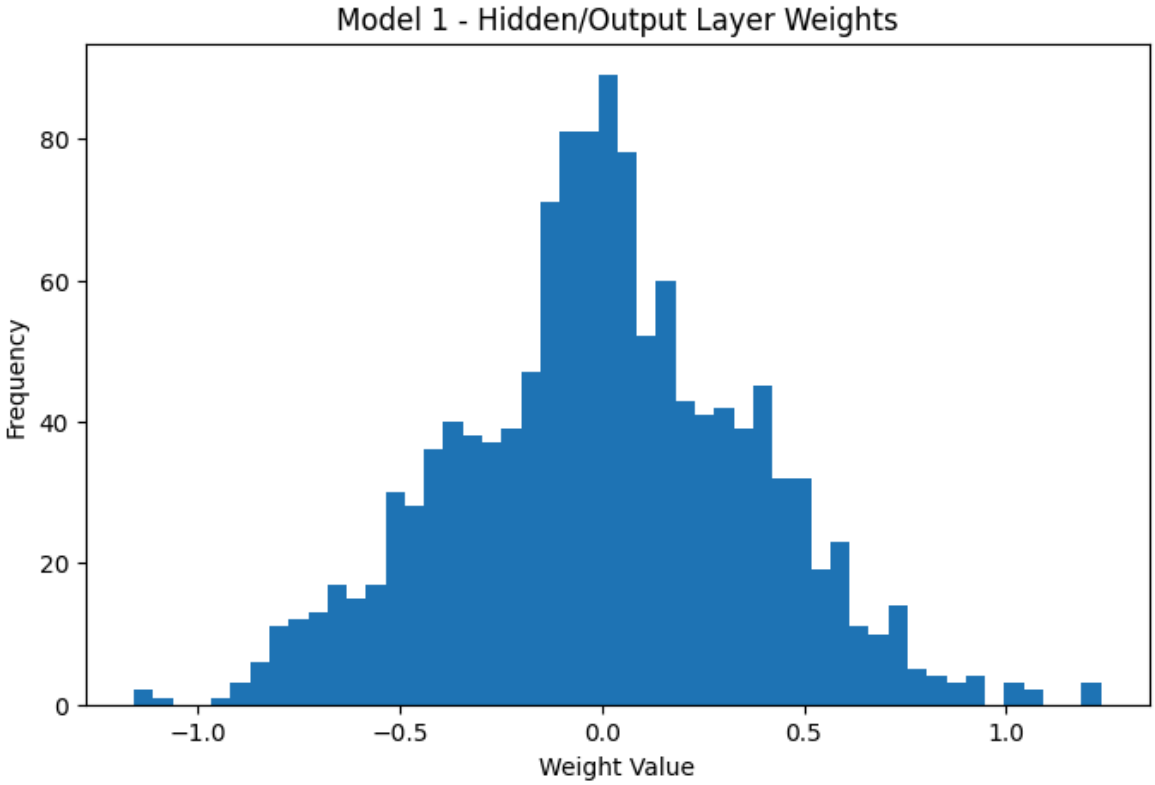
```
Model 1 - Epoch 31/40
Train Loss: 0.2186, Train Acc: 0.9197
Test  Loss: 0.3309, Test  Acc: 0.8855
----------------------------------------
Model 1 - Epoch 32/40
Train Loss: 0.2159, Train Acc: 0.9215
Test  Loss: 0.3174, Test  Acc: 0.8880
----------------------------------------
Model 1 - Epoch 33/40
Train Loss: 0.2124, Train Acc: 0.9223
Test  Loss: 0.3322, Test  Acc: 0.8835
----------------------------------------
Model 1 - Epoch 34/40
Train Loss: 0.2107, Train Acc: 0.9226
Test  Loss: 0.3226, Test  Acc: 0.8870
----------------------------------------
Model 1 - Epoch 35/40
Train Loss: 0.2070, Train Acc: 0.9255
Test  Loss: 0.3327, Test  Acc: 0.8863
----------------------------------------
Model 1 - Epoch 36/40
Train Loss: 0.2049, Train Acc: 0.9252
Test  Loss: 0.3338, Test  Acc: 0.8813
----------------------------------------
Model 1 - Epoch 37/40
Train Loss: 0.2015, Train Acc: 0.9263
Test  Loss: 0.3226, Test  Acc: 0.8864
----------------------------------------
Model 1 - Epoch 38/40
Train Loss: 0.2000, Train Acc: 0.9270
Test  Loss: 0.3247, Test  Acc: 0.8900
----------------------------------------
Model 1 - Epoch 39/40
Train Loss: 0.1976, Train Acc: 0.9296
Test  Loss: 0.3271, Test  Acc: 0.8841
----------------------------------------
Model 1 - Epoch 40/40
Train Loss: 0.1942, Train Acc: 0.9287
Test  Loss: 0.3307, Test  Acc: 0.8865
----------------------------------------
Model 2 - Epoch 1/40
Train Loss: 0.7634, Train Acc: 0.7335
Test  Loss: 0.5398, Test  Acc: 0.8089
----------------------------------------
Model 2 - Epoch 2/40
Train Loss: 0.5161, Train Acc: 0.8185
Test  Loss: 0.4863, Test  Acc: 0.8247
----------------------------------------
Model 2 - Epoch 3/40
Train Loss: 0.4682, Train Acc: 0.8351
Test  Loss: 0.4565, Test  Acc: 0.8332
----------------------------------------
Model 2 - Epoch 4/40
Train Loss: 0.4418, Train Acc: 0.8417
Test  Loss: 0.4253, Test  Acc: 0.8471
----------------------------------------
Model 2 - Epoch 5/40
Train Loss: 0.4263, Train Acc: 0.8484
Test  Loss: 0.4126, Test  Acc: 0.8496
----------------------------------------
```

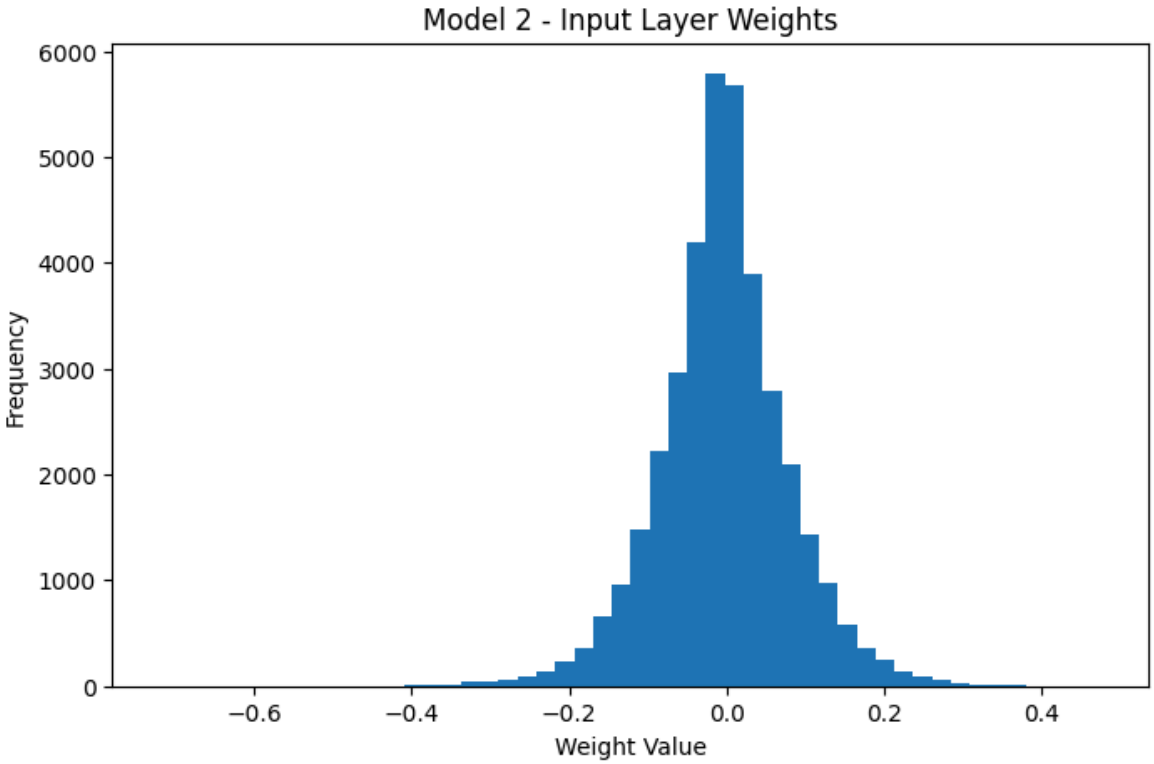
```
Model 2 - Epoch 6/40
Train Loss: 0.4110, Train Acc: 0.8525
Test  Loss: 0.4076, Test  Acc: 0.8519
----------------------------------------
Model 2 - Epoch 7/40
Train Loss: 0.4012, Train Acc: 0.8576
Test  Loss: 0.4066, Test  Acc: 0.8554
----------------------------------------
Model 2 - Epoch 8/40
Train Loss: 0.3936, Train Acc: 0.8595
Test  Loss: 0.4001, Test  Acc: 0.8530
----------------------------------------
Model 2 - Epoch 9/40
Train Loss: 0.3831, Train Acc: 0.8617
Test  Loss: 0.3886, Test  Acc: 0.8593
----------------------------------------
Model 2 - Epoch 10/40
Train Loss: 0.3758, Train Acc: 0.8665
Test  Loss: 0.3815, Test  Acc: 0.8657
----------------------------------------
Model 2 - Epoch 11/40
Train Loss: 0.3718, Train Acc: 0.8672
Test  Loss: 0.3737, Test  Acc: 0.8681
----------------------------------------
Model 2 - Epoch 12/40
Train Loss: 0.3655, Train Acc: 0.8694
Test  Loss: 0.3800, Test  Acc: 0.8618
----------------------------------------
Model 2 - Epoch 13/40
Train Loss: 0.3640, Train Acc: 0.8686
Test  Loss: 0.3709, Test  Acc: 0.8685
----------------------------------------
Model 2 - Epoch 14/40
Train Loss: 0.3589, Train Acc: 0.8716
Test  Loss: 0.3863, Test  Acc: 0.8573
----------------------------------------
Model 2 - Epoch 15/40
Train Loss: 0.3518, Train Acc: 0.8721
Test  Loss: 0.3667, Test  Acc: 0.8712
----------------------------------------
Model 2 - Epoch 16/40
Train Loss: 0.3502, Train Acc: 0.8734
Test  Loss: 0.3613, Test  Acc: 0.8725
----------------------------------------
Model 2 - Epoch 17/40
Train Loss: 0.3453, Train Acc: 0.8728
Test  Loss: 0.3659, Test  Acc: 0.8681
----------------------------------------
Model 2 - Epoch 18/40
Train Loss: 0.3451, Train Acc: 0.8749
Test  Loss: 0.3602, Test  Acc: 0.8732
----------------------------------------
Model 2 - Epoch 19/40
Train Loss: 0.3395, Train Acc: 0.8767
Test  Loss: 0.3707, Test  Acc: 0.8668
----------------------------------------
Model 2 - Epoch 20/40
Train Loss: 0.3399, Train Acc: 0.8767
Test  Loss: 0.3886, Test  Acc: 0.8609
----------------------------------------
```

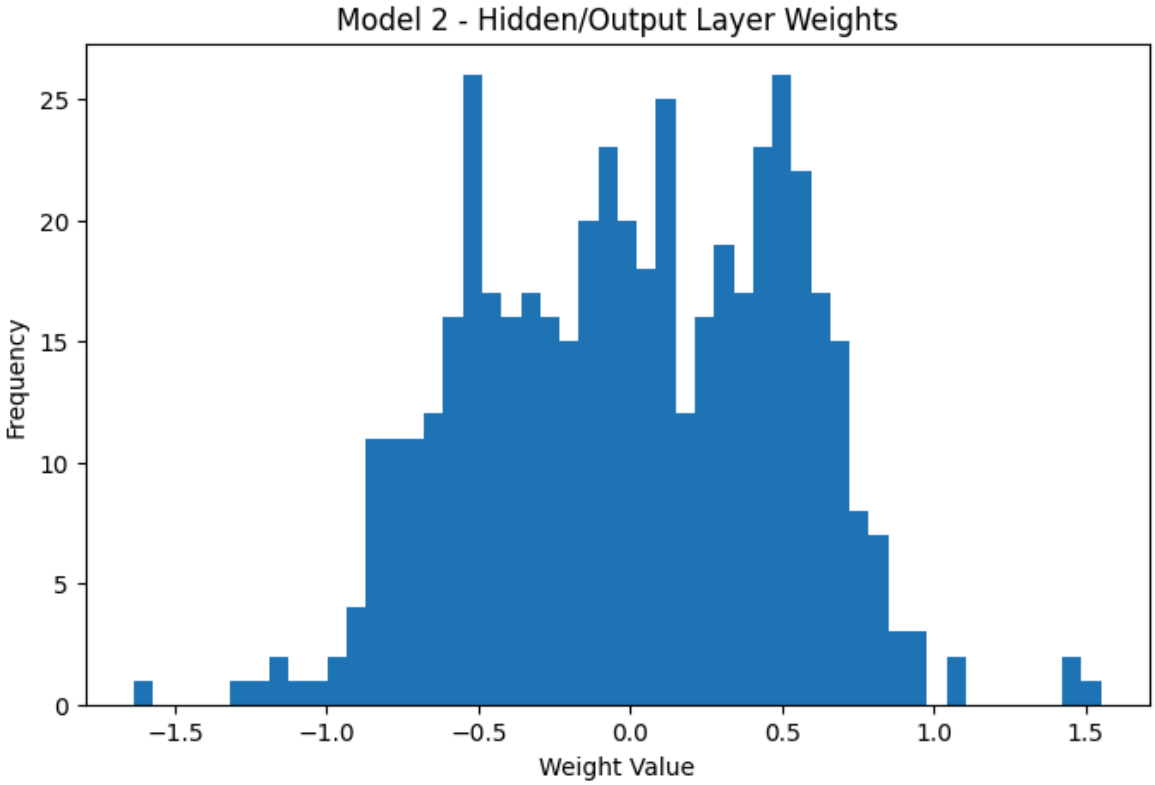
```
Model 2 - Epoch 21/40
Train Loss: 0.3359, Train Acc: 0.8786
Test  Loss: 0.3610, Test  Acc: 0.8708
----------------------------------------
Model 2 - Epoch 22/40
Train Loss: 0.3310, Train Acc: 0.8806
Test  Loss: 0.3651, Test  Acc: 0.8684
----------------------------------------
Model 2 - Epoch 23/40
Train Loss: 0.3309, Train Acc: 0.8810
Test  Loss: 0.3499, Test  Acc: 0.8748
----------------------------------------
Model 2 - Epoch 24/40
Train Loss: 0.3275, Train Acc: 0.8805
Test  Loss: 0.3591, Test  Acc: 0.8711
----------------------------------------
Model 2 - Epoch 25/40
Train Loss: 0.3260, Train Acc: 0.8811
Test  Loss: 0.3482, Test  Acc: 0.8758
----------------------------------------
Model 2 - Epoch 26/40
Train Loss: 0.3265, Train Acc: 0.8806
Test  Loss: 0.3517, Test  Acc: 0.8764
----------------------------------------
Model 2 - Epoch 27/40
Train Loss: 0.3192, Train Acc: 0.8837
Test  Loss: 0.3572, Test  Acc: 0.8756
----------------------------------------
Model 2 - Epoch 28/40
Train Loss: 0.3206, Train Acc: 0.8819
Test  Loss: 0.3511, Test  Acc: 0.8779
----------------------------------------
Model 2 - Epoch 29/40
Train Loss: 0.3194, Train Acc: 0.8835
Test  Loss: 0.3484, Test  Acc: 0.8761
----------------------------------------
Model 2 - Epoch 30/40
Train Loss: 0.3148, Train Acc: 0.8852
Test  Loss: 0.3515, Test  Acc: 0.8775
----------------------------------------
Model 2 - Epoch 31/40
Train Loss: 0.3146, Train Acc: 0.8844
Test  Loss: 0.3511, Test  Acc: 0.8772
----------------------------------------
Model 2 - Epoch 32/40
Train Loss: 0.3122, Train Acc: 0.8864
Test  Loss: 0.3457, Test  Acc: 0.8759
----------------------------------------
Model 2 - Epoch 33/40
Train Loss: 0.3110, Train Acc: 0.8858
Test  Loss: 0.3474, Test  Acc: 0.8785
----------------------------------------
Model 2 - Epoch 34/40
Train Loss: 0.3102, Train Acc: 0.8857
Test  Loss: 0.3473, Test  Acc: 0.8773
----------------------------------------
Model 2 - Epoch 35/40
Train Loss: 0.3082, Train Acc: 0.8867
Test  Loss: 0.3478, Test  Acc: 0.8787
----------------------------------------
```

```
Model 2 - Epoch 36/40
Train Loss: 0.3063, Train Acc: 0.8878
Test  Loss: 0.3508, Test  Acc: 0.8744
----------------------------------------
Model 2 - Epoch 37/40
Train Loss: 0.3042, Train Acc: 0.8879
Test  Loss: 0.3536, Test  Acc: 0.8763
----------------------------------------
Model 2 - Epoch 38/40
Train Loss: 0.3043, Train Acc: 0.8887
Test  Loss: 0.3511, Test  Acc: 0.8740
----------------------------------------
Model 2 - Epoch 39/40
Train Loss: 0.3051, Train Acc: 0.8872
Test  Loss: 0.3407, Test  Acc: 0.8802
----------------------------------------
Model 2 - Epoch 40/40
Train Loss: 0.3025, Train Acc: 0.8886
Test  Loss: 0.3463, Test  Acc: 0.8778
----------------------------------------
```




Model 1 - Hidden/Output Layer Weights
Frequency
Weight Value
0
20
40
60
80
−1.0
−0.5
0.0
0.5
1.0


Model 2 - Input Layer Weights
Frequency
Weight Value
0
1000
2000
3000
4000
5000
6000
−0.6
−0.4
−0.2
0.0
0.2
0.4



Model comparison.

Two neural networks were trained on Fashion MNIST for 40 epochs. Model 1 used one hidden layer with 128 ReLU units and did not use regularization or dropout. Model 2 used one hidden layer with 48 ReLU units, L2 regularization with coefficient 0.0001, and dropout with rate 0.2. Both models were trained using SGD and cross-entropy loss.

Histogram comparison

The weight histograms of Model 1 are more spread out, with a wider range of positive and negative values. In particular, the hidden/output layer of Model 1 contains more large-magnitude weights. In contrast, the weights of Model 2 are more concentrated around zero, especially in the input layer histogram. The distribution is narrower and contains fewer extreme values.

Effect of regularization

Regularization makes the weight distribution more concentrated around zero. It reduces the magnitude of the weights and discourages excessively large positive or negative values. As a result, the model becomes less complex and is less likely to overfit the training data. Therefore, regularization helps improve generalization by shrinking the weights toward zero.

In [ ]: